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Table Of Content

No	Table Of Content	Pages
1.	Introduction	3-4
2.	Review of Existing Solutions	5
3.	Methodology	6-10
4.	Results	11-12
5.	Prototype Showcase	13-19
6.	AI Acknowledgement	20
7.	Conclusion	21

Introduction

1.1 Background of Case Study

Digital financial services in ASEAN countries are growing rapidly as more people use mobile banking, digital payments, and online financial platforms. It is projected that digital payments in the ASEAN region will exceed $1.5 trillion by 2030. While these technologies increase access to financial services, many individuals still lack the financial knowledge required to manage these tools effectively. This creates a gap between the availability of financial services and users' understanding of financial risks, interest rates, credit management, and long-term financial planning.

In addition, many individuals in Southeast Asia face a significant financial protection gap. This means that people often do not have sufficient savings, insurance, or financial planning to cope with unexpected events such as medical emergencies, job loss, or economic downturns. Limited financial literacy makes it difficult for users to make informed financial decisions, which may lead to excessive debt, poor savings habits, and financial instability. Therefore, improving financial education and providing accessible financial guidance is important for strengthening financial resilience in the region.

1.2 Chosen SDG

This project supports **Sustainable Development Goal (SDG) 4: Quality Education**, which focuses on ensuring inclusive and equitable quality education and promoting lifelong learning opportunities for all. Financial literacy is an important form of practical education that helps individuals make responsible financial decisions throughout their lives. By leveraging artificial intelligence and data analysis, this project provides educational tools, personalized financial advice, and interactive financial simulations that help users better understand financial concepts such as savings, debt management, risk assessment, and financial planning.

Through these learning features, the system aims to improve financial knowledge among users and make financial education more accessible, especially for individuals who may not have formal financial training.

1.3 Problem Statement

Despite the rapid expansion of digital financial services in ASEAN countries, many individuals still lack sufficient financial literacy to effectively manage their finances. The widespread use of digital tools such as Buy Now Pay Later (BNPL), online lending platforms, and digital wallets has made financial services more accessible, but it has also increased the risk of poor financial decisions. Many users do not fully understand interest rates, debt obligations, or long-term financial consequences when using these services.

As a result, individuals may accumulate high levels of debt, have low savings, and lack adequate financial protection such as insurance or emergency funds. This situation creates financial vulnerability and limits individuals' ability to achieve financial stability. Therefore, there is a need for a system that can help users analyze their financial situation, identify potential risks, and provide guidance to improve their financial literacy and decision-making.

1.4 Objectives

The objective of this project is to develop an AI-powered financial literacy and risk assessment system that helps users better understand and manage their personal finances. The system analyzes users' financial data to evaluate their financial literacy levels and identify potential financial risks. It also integrates a machine learning model to predict users' financial risk levels based on financial indicators such as income, savings, credit score, and debt-to-income ratio.

In addition, the system provides an AI-based financial advisor that offers personalized financial recommendations based on users' profiles and financial goals. Interactive tools such as financial risk simulators and financial calculators are also included to help users learn important financial concepts in a practical way. Overall, the system aims to improve financial literacy, support better financial decision-making, and reduce the financial protection gap among individuals in the ASEAN region.

2. Review of Existing Solutions

Several financial technology (FinTech) applications and AI-powered financial advisory tools have been developed to help users manage their finances and improve financial literacy. These solutions provide features such as budgeting, savings management, investment advice, and financial data analysis.

One example is **MoneyWiz**, a personal finance application that allows users to track income, expenses, and budgets while generating financial reports to help users monitor their financial activities. The application helps individuals organize their finances and schedule payments more efficiently.

Another example is **Qapital**, a mobile financial management application that encourages users to save money through rule-based automation and gamification. The system automatically transfers money into savings accounts based on predefined rules, helping users develop better financial habits.

In addition, AI-powered financial tools such as robo-advisors and AI finance assistants have become increasingly popular. Platforms like Wealthfront and Albert use artificial intelligence to analyze users' financial data and provide personalized financial advice, budgeting suggestions, and investment recommendations. These tools help users make more informed financial decisions through automated financial analysis and guidance.

Although these existing solutions provide useful financial management features, several limitations still exist. Many financial applications focus primarily on budgeting or investment management rather than improving users' financial literacy. Some systems also require users to have prior financial knowledge in order to fully understand the advice provided. Additionally, certain tools may lack interactive educational features or personalized explanations that help users learn financial concepts step-by-step.

Therefore, the proposed system aims to address these gaps by integrating financial education, risk analysis, and AI-powered personalized recommendations into a single platform. The system not only analyzes users' financial data but also provides clear explanations, financial learning tools, and interactive simulations to help users better understand financial decisions and improve their financial literacy.

3. Methodology

3.1 System Architecture

The proposed system is an AI-powered financial literacy and risk assessment platform designed to help users understand their financial conditions and improve financial decision-making. The system integrates data analysis, machine learning, and artificial intelligence to provide personalized financial insights and educational tools.

The system architecture consists of four main components: data processing, machine learning risk prediction, AI financial advisor, and interactive financial education tools. The platform is implemented using Streamlit as the user interface, allowing users to interact with the system through a web-based dashboard.

First, the dataset containing synthetic personal financial data is loaded and processed using the Python Pandas library. This dataset includes financial attributes such as age, income, credit score, savings, and debt-to-income ratio. The system performs data preprocessing and calculates additional indicators such as financial literacy scores and financial protection gap scores.

Next, a machine learning model is trained to predict users' financial risk levels. A Random Forest classifier from the Scikit-learn library is used because it performs well with structured financial data and can capture nonlinear relationships between variables. The model analyzes multiple financial indicators and classifies users into different financial risk levels.

The system also integrates a generative AI model using the Google Gemini API. This AI component acts as a financial advisor that provides personalized financial recommendations based on the user's financial profile and financial goals.

Finally, the platform includes interactive financial learning tools such as financial risk simulators and financial calculators. These tools allow users to experiment with different financial scenarios and better understand financial concepts such as interest accumulation, debt management, and savings planning.

3.2 Technology Stack

The system is developed using several technologies that support data processing, machine learning, and interactive visualization.

- **Python** is used as the main programming language for data processing and system development.
- **Streamlit** is used to build the interactive web interface for the application.
- **Pandas and NumPy** are used for financial data processing and analysis.
- **Scikit-learn** is used to develop the machine learning model for financial risk prediction.
- **Plotly** is used to generate interactive charts and visualizations for financial data analysis.
- **Google Gemini API** is integrated to provide AI-powered financial advisory services.

These technologies work together to create an integrated financial literacy platform that combines analytics, artificial intelligence, and educational tools.

3.3 Machine Learning Risk Prediction Model

To evaluate users' financial risk levels, the system uses a Random Forest classification model. Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

The model uses several financial indicators as input features, including:

- Age
- Monthly income
- Credit score
- Debt-to-income ratio
- Savings amount

These features are used to train the model to classify users into different financial risk categories such as low risk, medium risk, and high risk.

The dataset is divided into training and testing sets using a train-test split approach. The model is trained on the training dataset and evaluated using the testing dataset to measure prediction accuracy.

Once the model is trained, it can be used to analyze a user's financial profile and predict the user's financial risk level. The result is displayed in the system dashboard along with recommendations for improving financial stability.

3.4 Financial Literacy and Protection Gap Assessment

In addition to machine learning prediction, the system calculates two additional indicators: financial literacy score and financial protection gap score.

The financial literacy score evaluates a user's understanding of financial management based on factors such as credit behavior, savings habits, and debt management. The score is calculated using a rule-based scoring method and ranges from 0 to 100.

The financial protection gap score measures the user’s level of financial preparedness in dealing with unexpected financial risks such as emergencies or income loss. This score considers factors such as insurance coverage, emergency savings, and income stability.

These indicators help users better understand their financial strengths and weaknesses while providing clear explanations to improve financial awareness.

3.5 Interactive Financial Learning Tools

To enhance financial education, the system includes several interactive learning tools.

One of the tools is a Buy Now Pay Later (BNPL) risk simulator that allows users to estimate the financial impact of installment-based purchases. Users can input purchase values and income levels to evaluate whether the installment payment is financially manageable.

Another feature is a compound interest calculator that demonstrates how savings or investments can grow over time. This helps users understand the long-term benefits of saving and investing.

These tools provide hands-on learning experiences that make financial concepts easier to understand and apply in real-life situations.

4. Results

4.1 System Implementation

The proposed AI-powered financial literacy platform was successfully implemented as a web-based application using Streamlit. The system integrates financial data analysis, machine learning risk prediction, and an AI-based financial advisor within a single platform.

Users can upload financial datasets, analyze financial literacy indicators, and interact with the system through various educational tools and dashboards. The system provides visualizations such as charts and graphs to help users better understand financial patterns and risk factors.

4.2 Machine Learning Prediction Results

The Random Forest model was successfully trained using the financial dataset and was able to classify users into different financial risk categories. Based on the evaluation using the testing dataset, the model demonstrated satisfactory prediction performance for identifying users with potential financial risk.

The model predictions allow the system to provide users with personalized financial insights and highlight potential financial vulnerabilities such as high debt levels or insufficient savings.

4.3 AI Financial Advisor

The integration of the Gemini AI model enables the system to provide personalized financial guidance. Users can interact with the AI advisor by asking financial-related questions or requesting advice based on their financial profiles.

The AI advisor analyzes user inputs together with financial indicators to generate recommendations related to budgeting, debt management, savings strategies, and financial planning.

4.4 Achievement of Project Objectives

Overall, the project successfully achieved its objectives of developing an AI-powered financial literacy and risk assessment system. The system combines data analysis, machine learning, and artificial intelligence to help users understand their financial situations and make better financial decisions.

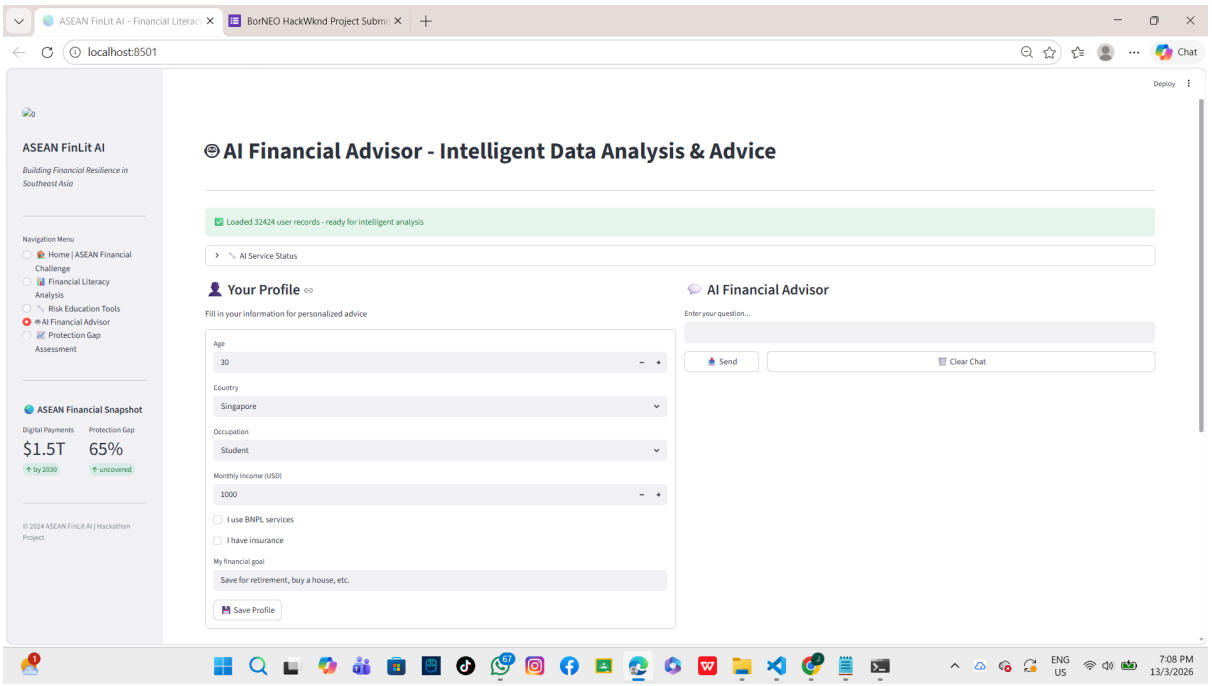
By providing financial education tools, personalized recommendations, and risk assessment features, the platform helps address financial literacy challenges and contributes to reducing the financial protection gap in the ASEAN region.

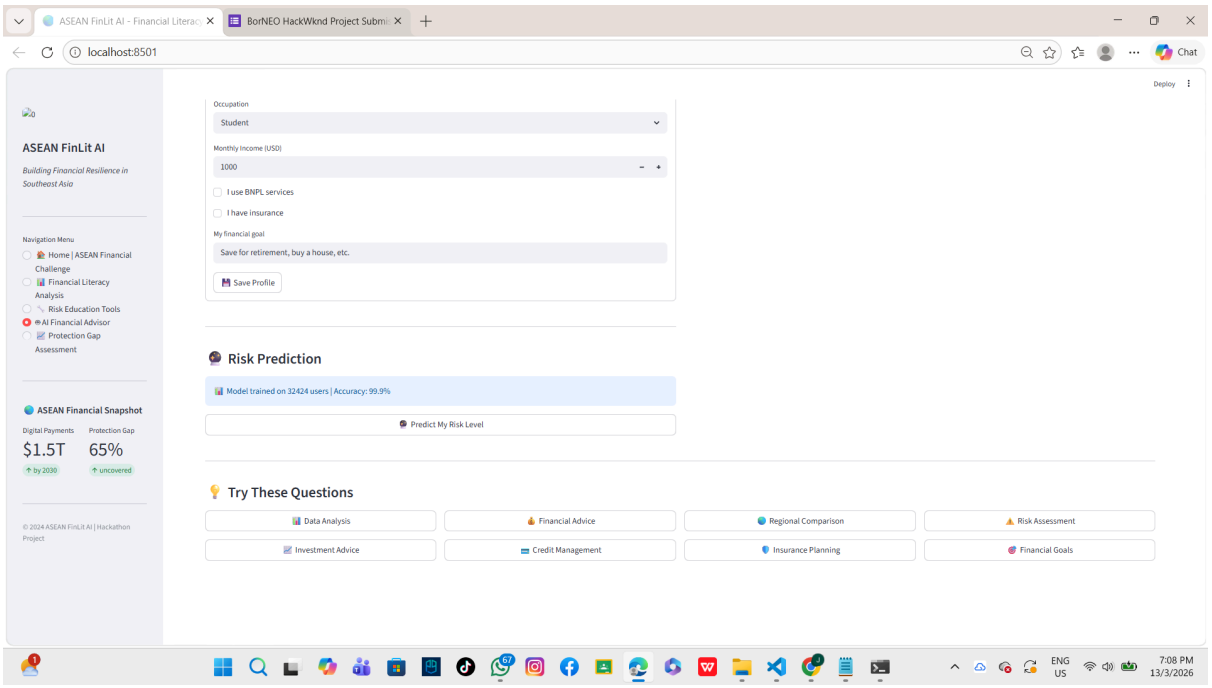
Prototype Showcase

User Interface Screenshot

Key features of the prototype include:

1. AI Financial Tutor Interface





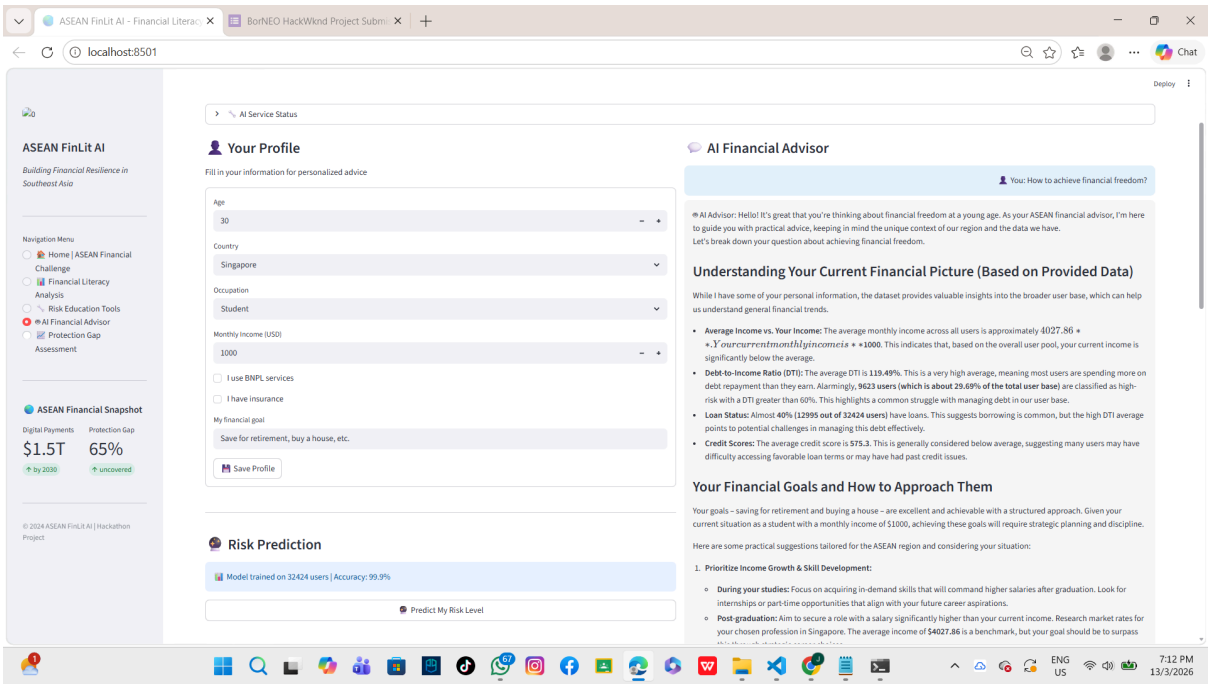
The **AI Financial Tutor Interface** allows users to interact with an AI assistant by asking financial-related questions. The system uses natural language processing to understand user queries and provide personalized financial guidance.

Users can ask questions such as:

- “Which insurance plan is suitable for me based on my financial situation?”
- “How much should I save every month?”
- “What type of financial protection should I consider?”

Based on the user’s financial information and context, the AI tutor provides simplified explanations, recommendations, and guidance to help users make informed financial decisions.

2. Financial Health Assessment Page



The **Financial Health Assessment Page** enables users to input their personal financial information so the system can evaluate their financial condition.This page is can based on financial information to give the better insurance and or financial goal and so on.

Users are required to provide details such as:

- **Age**
- **Occupation**

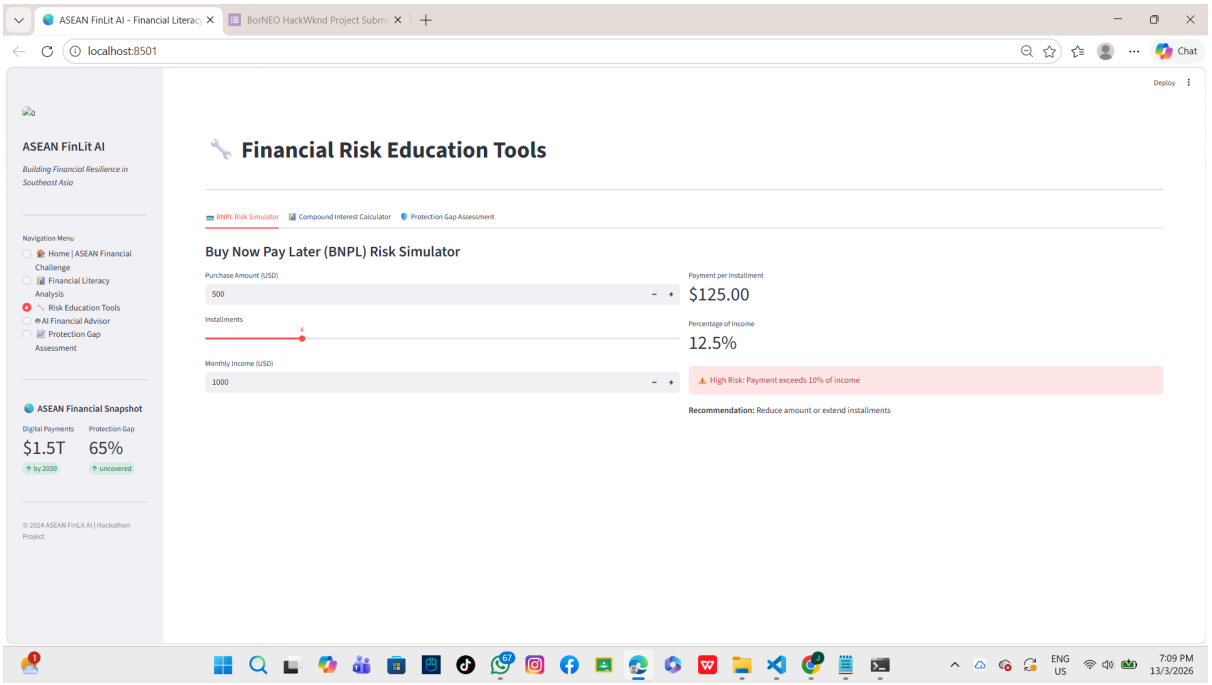
- **Monthly income**
- **Current insurance coverage**
- **Savings or financial commitments** (optional if included)

After analyzing the provided data, the system evaluates the user's financial condition and generates a **personalized financial goal and recommendation**. These goals may include suggestions such as:

- Recommended monthly savings targets
- Suggested emergency fund levels
- Suitable insurance protection
- Financial risk level assessment

This feature helps users better understand their financial status and provides actionable steps to improve their long-term financial stability.

3. Risk Prediction



The **Risk Prediction** module uses financial data provided by users to estimate their potential financial risk level. The system analyses factors such as:

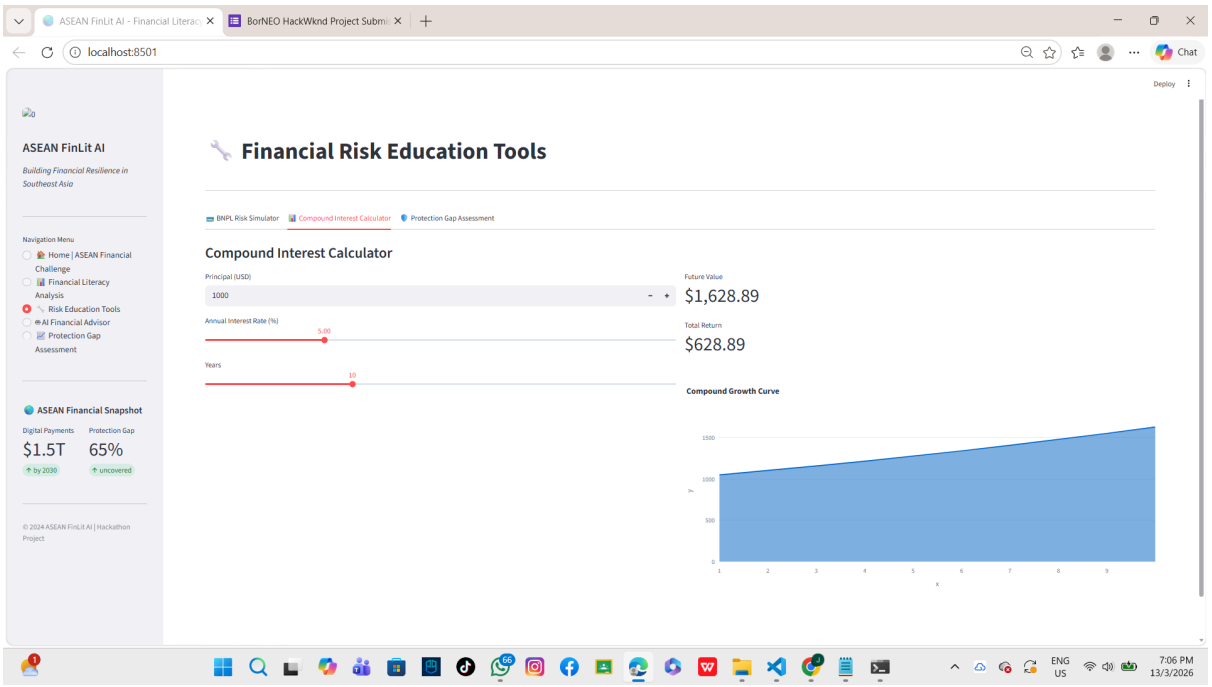
- Income stability
- Savings level
- Monthly expenses
- Debt obligations

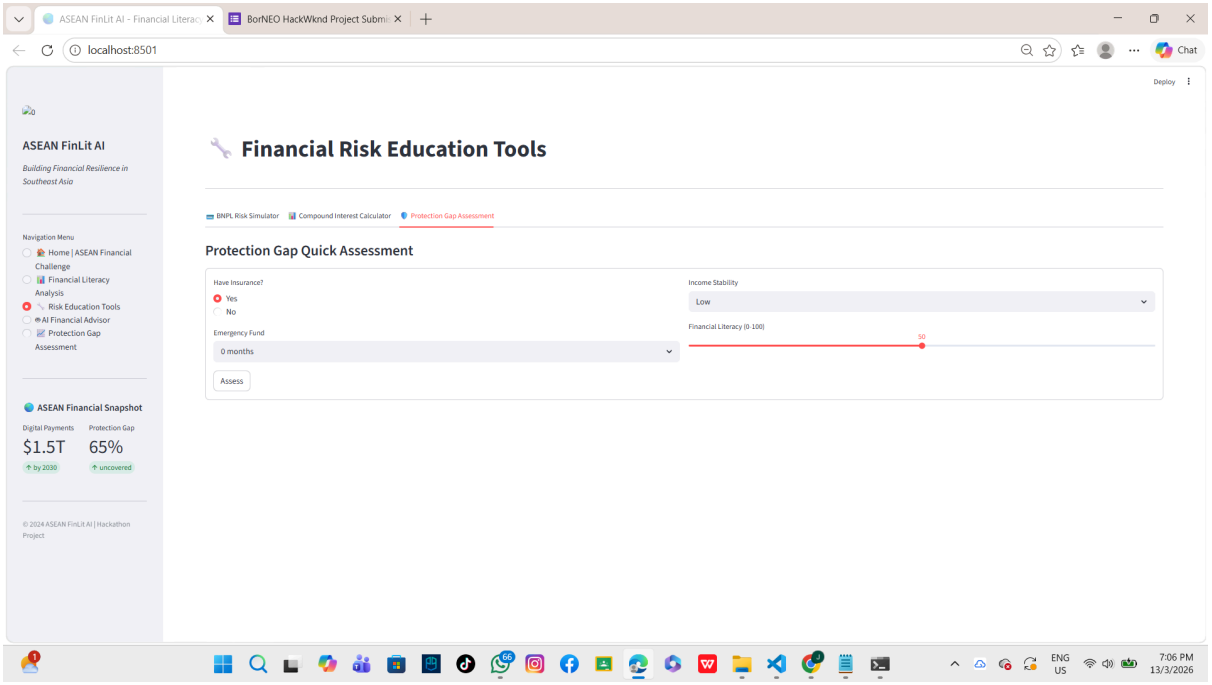
Using these indicators, the system predicts the user's financial risk category, such as:

- **Low Risk** – strong financial stability and adequate savings
- **Medium Risk** – manageable financial situation but improvements are needed
- **High Risk** – limited financial security and higher vulnerability to financial shocks

This prediction helps users recognize potential financial vulnerabilities and take preventive actions.

4. Risk Education Tools





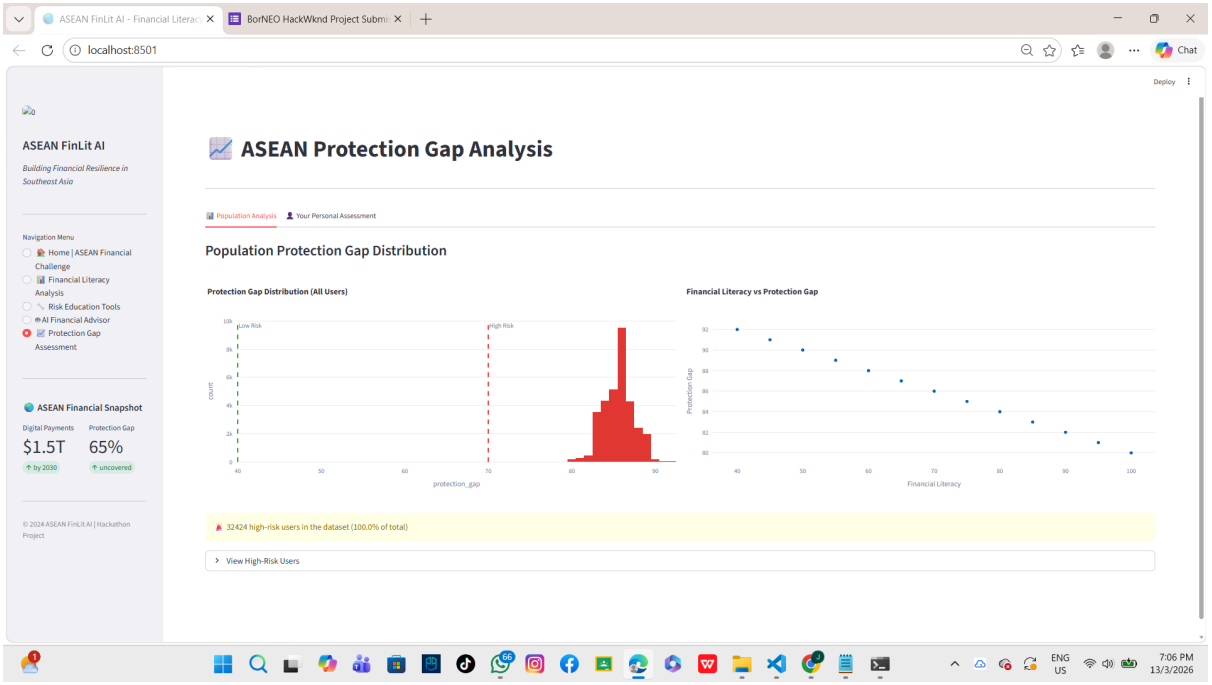
The **Risk Education Tools** component is designed to improve users' understanding of financial risks and protection strategies. Many individuals lack awareness of risks related to debt, insufficient savings, or lack of insurance coverage.

This feature provides educational content and interactive guidance that explains:

- Different types of financial risks
- The importance of emergency savings
- Insurance and financial protection strategies
- Long-term financial planning concepts

The AI system presents these topics in simplified and accessible language, helping users gain practical financial knowledge and build financial resilience.

5. Protection Gap Assessment



The **Protection Gap Assessment** feature identifies gaps in a user's financial protection, particularly in areas such as emergency funds, insurance coverage, and long-term financial security.

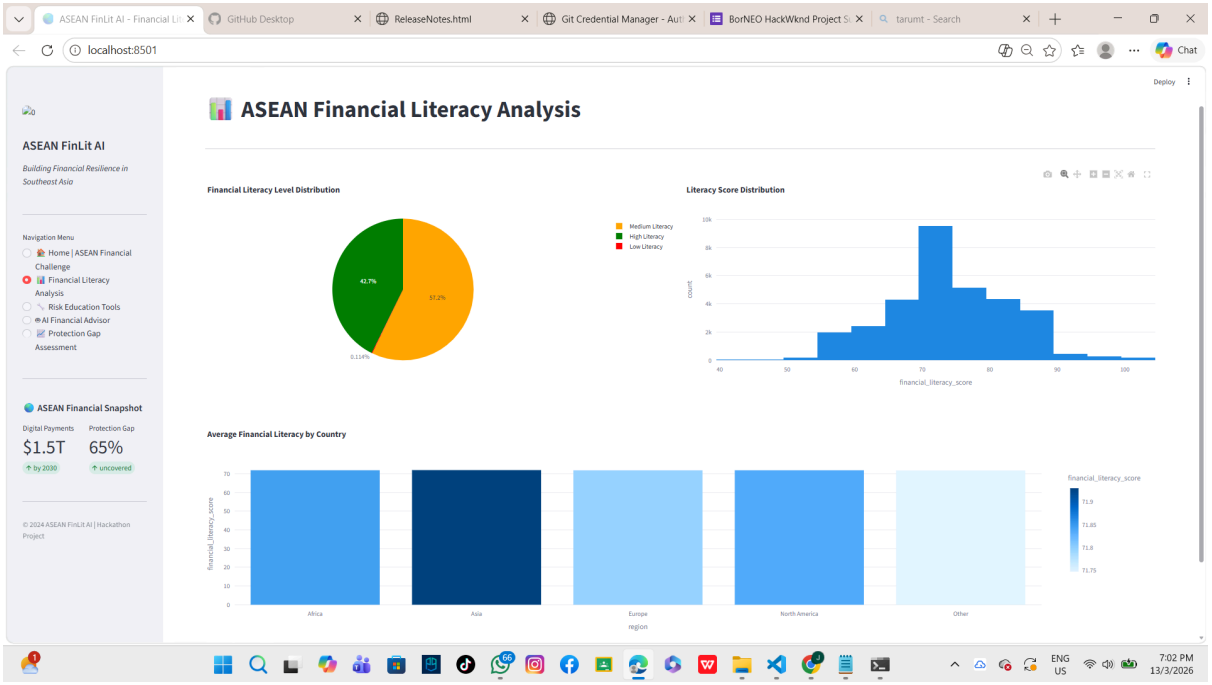
The system evaluates whether the user's current financial protection is sufficient to handle unexpected events such as:

- medical emergencies
- income loss
- economic downturns
- natural disasters

If a protection gap is detected, the system provides recommendations such as increasing emergency savings, considering suitable insurance options, or adjusting financial planning strategies.

This feature helps users strengthen their financial resilience and better prepare for unforeseen financial challenges.

6. Financial Literacy Analysis



The **Financial Literacy Analysis** module evaluates a user's financial knowledge and financial behavior through a set of structured inputs and indicators. Users provide information such as income, savings, expenses, and financial habits. The system then analyzes these inputs to determine the user's current level of financial literacy and financial management capability.

Based on the evaluation, the system provides personalized insights and recommendations to help users improve their financial planning, savings strategies, and spending habits. This feature aims to help users better understand their financial situation and develop stronger financial decision-making skills.

Acknowledgement

Disclosure

The following AI tools were used during the development of this project:

- ChatGPT
- Germini

Usage

These tools were used for:

- Assisting with **technical explanations and documentation**
- Supporting **code debugging and problem solving**
- Helping generate **structured report content and descriptions**

All final implementation, system design, and project decisions were completed by the project team.

Conclusion

This project presents an AI-powered platform designed to enhance financial literacy and provide inclusive financial guidance for individuals and micro-entrepreneurs in the ASEAN region. By integrating real-time data analysis, personalized recommendations, and educational tools, the system addresses the limitations of existing solutions, helping users make informed financial decisions, manage risks, and build long-term financial resilience.

For future improvements, the platform could incorporate multilingual support to reach a wider audience, include gamified learning modules to boost engagement, and integrate predictive analytics to forecast financial trends and provide proactive advice. Expanding partnerships with local financial institutions and community organizations could further enhance accessibility and ensure the platform reaches underserved communities effectively.